

# BUSFIN 8200: Problem Set 1

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## 1 Question 1

### 1.1 Part (a)

Assume  $D_t > 0$ . By definition, the gross return on equity is

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}. \quad (1)$$

Taking logs,

$$\begin{aligned} \log(R_{e,t+1}) &= \log\left(\frac{P_{t+1} + D_{t+1}}{P_t}\right) \\ &= \log\left(\frac{P_{t+1} + D_{t+1}}{P_{t+1}} \cdot P_{t+1}\right) - \log(P_t) \\ &= \log\left[\left(1 + \frac{D_{t+1}}{P_{t+1}}\right) P_{t+1}\right] - \log(P_t) \\ &= \log(P_{t+1}) + \log\left(1 + \frac{D_{t+1}}{P_{t+1}}\right) - \log(P_t). \end{aligned}$$

Define  $r_{e,t+1} \equiv \log(R_{e,t+1})$ ,  $p_t \equiv \log(P_t)$ , and  $d_t \equiv \log(D_t)$ . Since  $\frac{D_{t+1}}{P_{t+1}} = e^{\log(D_{t+1}/P_{t+1})} = e^{d_{t+1}-p_{t+1}}$ , we can write

$$r_{e,t+1} = p_{t+1} + \log\left(1 + e^{d_{t+1}-p_{t+1}}\right) - p_t.$$

Define  $f(x) \equiv \log(1 + e^x)$ , where  $x = d_{t+1} - p_{t+1}$ . We approximate this function around the average log dividend-price ratio  $\bar{d} - \bar{p}$  using a first-order Taylor approximation,

$$f(x) \approx f(\bar{d} - \bar{p}) + f'(\bar{d} - \bar{p})(x - \bar{d} - \bar{p}),$$

so that

$$f(d_{t+1} - p_{t+1}) \approx f(\bar{d} - \bar{p}) + f'(\bar{d} - \bar{p})[(d_{t+1} - p_{t+1}) - \bar{d} - \bar{p}].$$

Since  $f(x) = \log(1 + e^x)$ , we have  $f'(x) = \frac{e^x}{1 + e^x}$ , so

$$f'(\bar{d} - \bar{p}) = \frac{e^{\bar{d} - \bar{p}}}{1 + e^{\bar{d} - \bar{p}}}.$$

Define  $\kappa \equiv \frac{1}{1 + e^{\bar{d} - \bar{p}}}$ . Then  $1 - \kappa = \frac{e^{\bar{d} - \bar{p}}}{1 + e^{\bar{d} - \bar{p}}}$ , so  $f'(\bar{d} - \bar{p}) = 1 - \kappa$ .

Hence,

$$\begin{aligned} f(d_{t+1} - p_{t+1}) &\approx f(\overline{d-p}) + (1-\kappa) \left[ (d_{t+1} - p_{t+1}) - \overline{d-p} \right] \\ &= f(\overline{d-p}) - (1-\kappa)\overline{d-p} + (1-\kappa)(d_{t+1} - p_{t+1}). \end{aligned}$$

Define  $\kappa_0 \equiv f(\overline{d-p}) - (1-\kappa)\overline{d-p}$ , a constant. Since  $f(\overline{d-p}) = \log(1+e^{\overline{d-p}})$  and  $\kappa = \frac{1}{1+e^{\overline{d-p}}}$ , we have  $1+e^{\overline{d-p}} = \frac{1}{\kappa}$ , so

$$f(\overline{d-p}) = \log\left(\frac{1}{\kappa}\right) = -\log \kappa, \quad e^{\overline{d-p}} = \frac{1}{\kappa} - 1, \quad \overline{d-p} = \log\left(\frac{1}{\kappa} - 1\right).$$

Therefore,

$$\kappa_0 \equiv f(\overline{d-p}) - (1-\kappa)\overline{d-p} = -\log \kappa - (1-\kappa) \log\left(\frac{1}{\kappa} - 1\right).$$

We then have  $\log(1+e^{d_{t+1}-p_{t+1}}) = f(d_{t+1} - p_{t+1}) \approx \kappa_0 + (1-\kappa)(d_{t+1} - p_{t+1})$ , so

$$\begin{aligned} r_{e,t+1} &= p_{t+1} + \log\left(1+e^{d_{t+1}-p_{t+1}}\right) - p_t \\ &\approx p_{t+1} + \kappa_0 + (1-\kappa)(d_{t+1} - p_{t+1}) - p_t \\ &= \kappa_0 + (1-\kappa)d_{t+1} + \kappa p_{t+1} - p_t \\ &= \kappa_0 + \left[(1-\kappa)d_{t+1} - d_t + \kappa p_{t+1} - p_t + d_t\right] \\ &= \kappa_0 + (d_{t+1} - d_t) + \kappa(p_{t+1} - d_{t+1}) - (p_t - d_t) \\ &= \kappa_0 + \Delta d_{t+1} + \kappa(p_{t+1} - d_{t+1}) - (p_t - d_t). \end{aligned}$$

Therefore, we have

$$\begin{aligned} r_{e,t+1} &\approx \kappa_0 + \Delta d_{t+1} + \kappa(p_{t+1} - d_{t+1}) - (p_t - d_t), \\ \kappa &= \frac{1}{1+e^{\overline{d-p}}}, \\ \kappa_0 &= -\log(\kappa) - (1-\kappa) \log\left(\frac{1}{\kappa} - 1\right). \end{aligned}$$

Isolating  $p_t - d_t$  and recursively substituting for  $p_{t+h} - d_{t+h}$ , we have

$$\begin{aligned} (p_t - d_t) &\approx \kappa_0 - r_{e,t+1} + \Delta d_{t+1} + \kappa(p_{t+1} - d_{t+1}) \\ &= \kappa_0 - r_{e,t+1} + \Delta d_{t+1} + \kappa \left[ \kappa_0 - r_{e,t+2} + \Delta d_{t+2} + \kappa(p_{t+2} - d_{t+2}) \right] \\ &= (1+\kappa)\kappa_0 + (\Delta d_{t+1} + \kappa \Delta d_{t+2}) - (r_{e,t+1} + \kappa r_{e,t+2}) + \kappa^2(p_{t+2} - d_{t+2}) \\ &\vdots \text{ keep substituting } p_{t+h} - d_{t+h} \\ &= \frac{\kappa_0(1-\kappa^H)}{1-\kappa} + \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} - \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} + \kappa^H(p_{t+H} - d_{t+H}). \end{aligned}$$

Thus, we have

$$d_t - p_t \approx \frac{\kappa_0(\kappa^H - 1)}{1-\kappa} + \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H(d_{t+H} - p_{t+H}),$$

so, letting  $dp_t \equiv d_t - p_t$ ,

$$dp_t \approx \frac{\kappa_0(\kappa^H - 1)}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H dp_{t+H}. \quad (1.1)$$

Taking expectations as of time  $t$ ,

$$dp_t = \frac{\kappa_0(\kappa^H - 1)}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] + \kappa^H \mathbb{E}_t[dp_{t+H}]. \quad (1.2)$$

Suppose  $H \rightarrow \infty$ . Since  $\kappa = \frac{1}{1 + e^{\bar{d}-p}} \in (0, 1)$ , we have  $\lim_{H \rightarrow \infty} \kappa^H = 0$ , and by the no-bubble condition  $\lim_{H \rightarrow \infty} \kappa^H \mathbb{E}_t[dp_{t+H}] = 0$ . Therefore,

$$dp_t = \frac{-\kappa_0}{1 - \kappa} + \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}]. \quad (1.3)$$

## 1.2 Part (b)



Figure 1: OLS slope coefficients of `future_return`, `change_d`, and `dp_H` on `dp`, plotted against the horizon  $H$  (in years).

From the equation

$$1 = \underbrace{\frac{\text{Cov} \left[ dp_t, \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} \right]}{\text{Var}[dp]}}_{b_{re}^{(H)}} + \underbrace{\frac{\text{Cov} \left[ dp_t, -\sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} \right]}{\text{Var}[dp]}}_{b_{\Delta d}^{(H)}} + \underbrace{\frac{\text{Cov} [dp_t, \kappa^H dp_{t+H}]}{\text{Var}[dp]}}_{b_{dp}^{(H)}}.$$

This equation is trying to measure, how much each component moves when today's  $dp_t$  moves. The meaning of these three coefficients is as follows:

$$\sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] = a + b_{r_e}^{(H)} dp_t + \epsilon_t$$

The slope is  $b_{r_e}^{(H)}$ , and it measures the sensitivity of expected discounted future returns to today's dividend-price ratio.

Because dividend growth enters the Campbell–Shiller equation with a minus sign, define

$$-\sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] = a + b_{\Delta d}^{(H)} dp_t + \epsilon_t.$$

So  $b_{\Delta d}^{(H)}$  captures how much variation in  $dp_t$  is associated with expected future dividend growth.

$$\kappa^H \mathbb{E}_t[dp_{t+H}] = a + b_{dp}^{(H)} dp_t + \epsilon_t.$$

$b_{dp}^{(H)}$  tells us how much today's high  $dp_t$  is simply expected to persist into the future.

$$1 = b_{r_e}^{(H)} + b_{\Delta d}^{(H)} + b_{dp}^{(H)}$$

From the graph, we can see that the contributions of expected future returns, expected future dividend growth, and the future dividend-price ratio to the variation in today's  $dp_t$  change with the horizon  $H$ . In the short run, the variation in today's  $dp_t$  is primarily driven by the future dividend-price ratio. As the horizon  $H$  increases, the contribution of expected future returns becomes more significant, while the contribution of the future dividend-price ratio declines toward zero. The contribution of negative expected future dividend growth remains relatively stable across different horizons. Also, if we add up these three coefficients, we get approximately 1, which reflects the normalized variance decomposition of today's  $dp_t$ .

### 1.3 Part (c)



Figure 2: VAR-implied coefficients

The VAR-implied coefficients shown in Figure 2 show some interesting patterns. Instead of the expected future return component alone becoming dominant as  $H$  increases, the figure shows that both the expected future return component and the negative expected future dividend growth component contribute significantly to the variation in today's  $dp_t$  across different horizons. At the same time, the contribution of the future dividend-price ratio declines as the horizon increases, consistent with the findings from the direct-regression estimates, although the decline appears to be more gradual under the VAR approach. The difference in these patterns may arise because the two approaches estimate long-horizon coefficients differently. The direct approach estimates each horizon separately using realized future outcomes, whereas the VAR approach estimates the joint one-year dynamics of the state variables and uses the estimated transition matrix to imply coefficients at all subsequent horizons.

#### 1.4 Part (d)

The resulting long-run VAR-implied coefficients are reported in Table 1. The table shows that, in the long run, the contributions of expected future returns and negative expected future dividend growth are roughly equal, accounting for approximately 49% and 51%, respectively. Their sum is 0.998808, which is close to, but not exactly, one. This indicates that the unrestricted estimated VAR approximately, rather than mechanically, satisfies the long-run Campbell–Shiller restriction.

Table 1: Long-run VAR-implied coefficients

| Component                         | Coefficient                             | Estimate |
|-----------------------------------|-----------------------------------------|----------|
| Expected return                   | $b_{re}^{(\infty)}$                     | 0.489191 |
| Negative expected dividend growth | $b_{dg}^{(\infty)}$                     | 0.509617 |
| Sum                               | $b_{re}^{(\infty)} + b_{dg}^{(\infty)}$ | 0.998808 |

## 1.5 Part (e)

First, assume  $D_t > 0$ . We have

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}.$$

Then,

$$\begin{aligned} R_{e,t+1} \frac{D_t}{D_{t+1}} &= \frac{P_{t+1} + D_{t+1}}{P_t} \frac{D_t}{D_{t+1}} \\ &= \frac{D_t}{P_t} \frac{P_{t+1} + D_{t+1}}{D_{t+1}} \\ &= \frac{P_t + D_t}{P_t} \frac{D_t}{P_t + D_t} \\ &= \frac{D_t}{P_{t+1} + D_{t+1}}. \end{aligned}$$

Taking logs on both sides gives

$$\begin{aligned} \log(R_{e,t+1}) + \log\left(\frac{D_t}{D_{t+1}}\right) &= \log\left(\frac{P_t + D_t}{P_t}\right) + \log\left(\frac{D_t}{P_t + D_t}\right) \\ &\quad - \log\left(\frac{D_{t+1}}{P_{t+1} + D_{t+1}}\right). \end{aligned}$$

Define

$$r_{e,t+1} = \log(R_{e,t+1}), \quad p_t = \log(P_t), \quad d_t = \log(D_t), \quad dy_t = \log\left(1 + \frac{D_t}{P_t}\right).$$

Given that

$$\begin{aligned} \log\left(\frac{D_t}{P_t + D_t}\right) &= \log\left(1 - \frac{P_t}{P_t + D_t}\right) \\ &= \log\left(1 - e^{\log(P_t)/(P_t + D_t)}\right) \\ &= \log\left(1 - e^{-\log((P_t + D_t)/P_t)}\right) \\ &= \log\left(1 - e^{-\log(1 + D_t/P_t)}\right) \\ &= \log\left(1 - e^{-dy_t}\right), \end{aligned}$$

we have

$$\log\left(\frac{D_t}{P_t + D_t}\right) = \log\left(1 - e^{-dy_t}\right).$$

Similarly,

$$\log\left(\frac{D_{t+1}}{P_{t+1} + D_{t+1}}\right) = \log\left(1 - e^{-dy_{t+1}}\right).$$

Therefore,

$$r_{e,t+1} - \Delta d_{t+1} = dy_t + \log\left(1 - e^{-dy_t}\right) - \log\left(1 - e^{-dy_{t+1}}\right).$$

Let

$$f(x) = \log(1 - e^{-x}).$$

Then,

$$f'(x) = \frac{1}{1 - e^{-x}} e^{-x} = \frac{e^{-x}}{1 - e^{-x}}.$$

Let  $x = dy_t$ . By a first-order approximation, and defining  $\kappa \equiv e^{-\bar{dy}}$ , we have

$$\begin{aligned} \log(1 - e^{-dy_t}) &\approx \log(1 - e^{-\bar{dy}}) + \log'(1 - e^{-\bar{dy}})(dy_t - \bar{dy}) \\ &= \log(1 - \kappa) + \frac{e^{-\bar{dy}}}{1 - e^{-\bar{dy}}}(dy_t - \bar{dy}) \\ &= \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_t - \bar{dy}). \end{aligned}$$

Thus,

$$\log(1 - e^{-dy_t}) \approx \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_t - \bar{dy}).$$

Similarly,

$$\log(1 - e^{-dy_{t+1}}) \approx \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_{t+1} - \bar{dy}).$$

Substituting these approximations into  $r_{e,t+1} - \Delta d_{t+1} = dy_t + \log(1 - e^{-dy_t}) - \log(1 - e^{-dy_{t+1}})$ , we obtain

$$\begin{aligned} r_{e,t+1} - \Delta d_{t+1} &= dy_t + \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_t - \bar{dy}) \\ &\quad - \log(1 - \kappa) - \frac{\kappa}{1 - \kappa}(dy_{t+1} - \bar{dy}) \\ &= dy_t + \frac{\kappa}{1 - \kappa}(dy_t - dy_{t+1}) \\ &= \frac{1}{1 - \kappa}dy_t - \frac{\kappa}{1 - \kappa}dy_{t+1}. \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{1}{1 - \kappa}dy_t &= r_{e,t+1} - \Delta d_{t+1} + \frac{\kappa}{1 - \kappa}dy_{t+1}, \\ dy_t &= (1 - \kappa)(r_{e,t+1} - \Delta d_{t+1}) + \kappa dy_{t+1}. \end{aligned}$$

Recursively substituting for future values of  $dy$ , we have

$$\begin{aligned} dy_t &= (1 - \kappa)(r_{e,t+1} - \Delta d_{t+1}) \\ &\quad + \kappa[(1 - \kappa)(r_{e,t+2} - \Delta d_{t+2}) + \kappa dy_{t+2}] \\ &\quad \vdots \quad \text{keep substituting } dy_{t+h} \\ &= (1 - \kappa) \left( \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} \right) + \kappa^H dy_{t+H}. \end{aligned}$$

Thus, we have

$$dy_t = (1 - \kappa) \left( \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} \right) + \kappa^H dy_{t+H}.$$

Taking expectations at time  $t$ , we have

$$dy_t = (1 - \kappa) \left( \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] \right) + \kappa^H \mathbb{E}_t[dy_{t+H}].$$

Suppose  $H \rightarrow \infty$ . Given that  $\kappa = e^{-\bar{d}y} \in (0, 1)$ , we have

$$\lim_{H \rightarrow \infty} \kappa^H = 0.$$

Also, because of the no-bubble condition,

$$\lim_{H \rightarrow \infty} \kappa^H \mathbb{E}_t[dy_{t+H}] = 0.$$

Therefore,

$$dy_t = (1 - \kappa) \left( \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] \right).$$

## 2 Question 2

### 2.1 Part (a)

Figure 3 shows that the adjusted  $R^2$  generally rises with the horizon, from about 4.6% at one year to a peak of about 44.3% at 14 years, before declining slightly to about 42.1% at 15 years. Thus, the current dividend-price ratio has substantially greater explanatory power for long-horizon average excess returns than for one-year-ahead excess returns.



Figure 3: Adjusted  $R^2$  from predictive regressions of average future annual simple excess returns on the starting dividend-price ratio.

3 Question 3

4 Question 4